TENNESSEE
VALLEY
AUTHORITY

Post Office Box 2000, Decatur, Alabama 35609-2000

October 28, 2024

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Units 1, 2, and 3
Renewed Facility Operating License Nos. DPR-33, -52, and -68
NRC Docket Nos. 50-259, -260, and -296

Subject: **Licensee Event Report 50-260/2024-001-02**

References:
1. Letter from TVA to NRC, "Licensee Event Report 50-260/2024-001-00," dated February 16, 2024 (ML24047A222)
2. Letter from TVA to NRC, "Licensee Event Report 50-260/2024-001-01," dated April 17, 2024 (ML24108A183)

The enclosed Licensee Event Report provides details of a failed Secondary Containment Isolation Valve on Browns Ferry Nuclear Plant, Units 1, 2, and 3. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's Technical Specifications. Revision 2 to this report includes applicability to BFN Units 1 and 3 which was not previously reported.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David J. Renn, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,

Daniel A. Komm
Site Vice President

Enclosure: Licensee Event Report 50-260/2024-001-02 – Secondary Containment Isolation Valve Inoperable due to Mechanical Failure cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

ENCLOSURE

Browns Ferry Nuclear Plant
Units 1, 2, and 3

Licensee Event Report 50-260/2024-001-02

Secondary Containment Isolation Valve Inoperable due to Mechanical Failure

---

See Enclosed

NRC FORM 366 (04-02-2024)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 04/30/2027

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		52. Docket Number	3. Page
Browns Ferry Nuclear Plant, Unit 2	☒ 050 ☐ 052	00260	1 OF 7

4. Title

Secondary Containment Isolation Valve Inoperable due to Mechanical Failure

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
12	18	2023	2024	- 001 -	02	10	28	2024	Browns Ferry Nuclear Plant, Unit 1	05000259
									Browns Ferry Nuclear Plant, Unit 3	05000296

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Baruch Calkin, Licensing Engineer	256-278-1031

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
D	064	DMP	A613	No	N/A	N/A	N/A	N/A	N/A

4. Supplemental Report Expected)		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date		N/A	N/A	N/A

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On November 18, 2023, at approximately 2038 Central Standard Time, during performance of Surveillance Requirement 3.6.4.2.1, Secondary Containment Isolation Valve (SCIV) Stroke Timing, Reactor Zone Outboard Supply Damper 2-DMP-0064-0013 failed to close upon securing Reactor Zone Supply Fans. Operations personnel declared 2-DMP-064-0013 inoperable for SCIV function and entered Technical Specification Limiting Condition for Operation 3.6.4.2 Condition A. On November 22, 2023, 1515 CST, following corrective maintenance on the damper, Operations personnel declared damper 2-DMP-064-0013 operable for SCIV function and exited TS LCO 3.6.4.2 Condition A. On December 18, 2023, a Past Operability Evaluation determined that the damper had been inoperable for longer than allowed by TS. This condition did not result in any loss of safety function.

The apparent cause of this event was a lack of proper guidance in damper maintenance instructions. The corrective actions to reduce the possibility of recurrence for this condition are to replace broken and/or degraded components on the affected valve, and to revise maintenance procedures for the dampers to include an updated Preventative Maintenance plan with Electric Power Research Institute maintenance guidance. Revision 2 to this report includes applicability to BFN Units 1 and 3 which was not previously reported.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 2	☒ 050 / ☐ 052	00260	2024	- 001	- 02

NARRATIVE

I. Plant Operating Conditions before the Event

At the time of discovery, Browns Ferry Nuclear Plant (BFN) Units 1, 2, and 3 were each in Mode 1 at approximately 100 percent power.

II. Description of Event

A. Event Summary

On November 18, 2023, at approximately 2038 Central Standard Time (CST), during performance of Surveillance Requirement (SR) 3.6.4.2.1, Secondary Containment [JM] Isolation Valve (SCIV) [ISV] Stroke Timing, Reactor Zone Outboard Supply Damper [DMP] 2-DMP-064-0013 failed to close upon securing Reactor Zone Supply Fans [FAN]. Operations personnel declared 2-DMP-064-0013 inoperable for SCIV function and entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.6.4.2 Condition A. On November 22, 2023, 1515 CST, following implementation of a Temporary Modification (TMOD) to the damper, Operations personnel declared damper 2-DMP-064-0013 operable for SCIV function and exited TS LCO 3.6.4.2 Condition A.

On December 18, 2023, TVA completed a Past Operability Evaluation (POE) which determined that the damper had been inoperable for longer than allowed by TS.

The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's TS.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

NARRATIVE

C. Dates and approximate times of occurrences

DATE AND APPROXIMATE TIMES	OCCURRENCE
August 20, 2023 0015 CST	Declared 2-DMP-064-0013 inoperable for SCIV function. Declared operable following corrective maintenance.
September 24, 2023 2230 CST	Declared 2-DMP-064-0013 inoperable for SCIV function. Declared operable following corrective maintenance.
November 18, 2023 2038 CST	While performing 0-SR-3.6.4.2.1, Reactor Zone Outboard Supply Damper 2-DMP-064-0013 failed to close upon securing Reactor Zone Supply Fans. Declared 2-DMP-064-0013 inoperable for SCIV function. Entered TS LCO 3.6.4.2 Condition A, Actions A.1 and A.2
November 22, 2023 1515 CST	Declared 2-DMP-064-0013 operable for SCIV function following completion of corrective maintenance and PMT. Exited TS LCO 3.6.4.2 Condition A
December 18, 2023	Upon completion of POE, determined that 2-DMP-064-0013 had been inoperable longer than allowed by TS.

D. Manufacturer and model number of each component that failed during the event

The damper is part of a solenoid [SOL] controlled isolation valve made by Automatic Valve Corporation, part number U0203GBBR-AA.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

Subsequent troubleshooting determined that the mechanistic cause of failure was wear on the damper louver [LV] arm pin holes.

NARRATIVE

G. The failure mode, mechanism, and effect of each failed component

Wear on the damper louver arm pin holes allowed the louver arm to become loose and ultimately jam on top of the louver bracket, preventing closure.

H. Operator actions

There were no operator actions associated with this event.

I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this event.

III. Cause of the event

The apparent cause of this event was a lack of proper guidance in maintenance procedure MCI0000DMP001, Damper Maintenance Instructions.

There is no guidance in this procedure which would have driven maintenance to look at the component which failed. The degraded pinholes are hidden inside the ductwork of the damper and are partly covered up by the pin itself on the brackets. The pin hole on the louver arm is completely hidden inside the bracket. Without guidance to specifically look at this location, the condition would never be noticed. Electric Power Research Institute (EPRI) provides guidance on maintenance for couplings and connections on dampers which addresses the issue, but this guidance was not included in the damper maintenance instructions.

A. Cause of each component or system failure or personnel error

The mechanistic cause of failure was determined to be wear on the damper louver arm pin holes which allowed the louver arm to become loose and ultimately jam on top of the louver bracket, preventing closure.

B. Cause(s) and circumstances for each human performance related root cause

No human performance related root cause was identified.

NARRATIVE

IV. Analysis of the event

The secondary containment isolation is initiated from any of three signals: low reactor water level, high drywell pressure or high activity in a ventilation exhaust duct [DUCT], or by manual alignment and operation from the Main Control Room. Each signal simultaneously isolates the secondary containment zone or zones, shuts down normal ventilation equipment, opens dampers to and from the Standby Gas Treatment System [BH] and starts the Standby Gas Treatment System blower.

BFN Unit 2 TS LCO 3.6.4.2 states that each SCIV shall be Operable in Modes 1, 2, and 3. TS LCO 3.6.4.2 Actions A.1 and A.2 require, when one or more penetration [PEN] flow paths contain an Inoperable SCIV, that each affected penetration flow path be isolated by use of at least one closed and de-activated automatic valve, closed manual valve, or blind flange, within eight hours; and that isolation of each affected penetration flow path be verified every thirty-one days. A POE determined that the Reactor Zone Air Supply Outboard Isolation Valve 2-DMP-064-0013 was inoperable for longer than allowed by TS LCO 3.6.4.2.

V. Assessment of Safety Consequences

The redundant Inboard Isolation Valve 2-DMP-064-0014 remained operable during the period of time when the Outboard Isolation Valve 2-DMP-064-0013 was inoperable, maintaining secondary containment. Furthermore, Probabilistic Risk Assessment (PRA) conducted by TVA determined that the reactor zone air supply, including damper 2-DMP-064-0013, is not credited in the internal events/internal flooding, seismic, or Fire PRA Models.

Based on the above, the TVA has concluded that sufficient systems were available to provide the required safety functions needed to protect the health and safety of the public.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

The redundant Inboard Isolation Valve 2-DMP-064-0014 remained operable during this event, and there was no known inoperability of this valve during the period where 2-DMP-064-0013 was determined to be inoperable. Therefore, the secondary containment safety function was maintained during that time.

NARRATIVE

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

The Reactor Zone Air Supply Outboard Isolation Valve 2-DMP-064-0013 was inoperable from approximately 0015 CST on August 20, 2023 until approximately 1515 on November 22, 2023, approximately ninety-five days.

VI. Corrective Actions

Corrective Actions are being managed by the TVA's corrective action program under CR 1893273.

A. Immediate Corrective Actions

A temporary modification was implemented to pin the lower louvers closed until the valve was repaired.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

The corrective actions to reduce the possibility of recurrence for this condition are as follows:

1. Replace broken and/or degraded components on the affected valve.
2. Revise maintenance procedures associated with 1/2/3-DMP-064-0005/6 Refuel Zone Air Supply Isolation Dampers and 1/2/3-DMP-0013/14 Reactor Zone Air Supply Isolation Dampers to include an updated Preventative Maintenance plan with EPRI maintenance guidance.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 2	☒ 050 / ☐ 052	00260	2024	- 001	- 02

NARRATIVE

VII. Previous Similar Events at the Same Site

A search of LERs from BFN, Units 1, 2, and 3 over the last five years identified no similar events.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.